



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Generative AI for SEO

Course Code: C11

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 15.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
13.0	Not stated in supplied v13 deck	Legacy SEO for Small and Medium Enterprises reference deck.	Tertiary Infotech Academy Pte Ltd
14.0	16 August 2026	Generative-AI-assisted SEO, prompt frameworks, AI Overviews, multimodal content, agentic governance and ten progressive website labs.	Tertiary Infotech Academy Pte Ltd
15.0	17 August 2026	Added the eight field-tested meta-tag and LinkedIn prompt templates, the five-part prompt skeleton and the pre-publish checklist to the activity prompt packs.	Tertiary Infotech Academy Pte Ltd

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Course Overview

A one-day applied course on using generative AI, structured prompts and governed workflows to research, create, optimise, distribute and measure SEO content across classic and generative search experiences.

Learning Outcomes

- LO1: Evaluate SEO and internet marketing strategies of a website.
- LO2: Manage keyword research in alignment with SEO objectives.
- LO3: Provide on-page SEO strategies and recommendations.
- LO4: Manage off-page SEO strategies and performance across multiple channels.
- LO5: Monitor Google Analytics and Search Console reports to evaluate SEO performance.

Daily Schedule

Course Day

Time	Duration	Topic / Lab	Method	Slides
9:30 AM–10:00 AM	30 min	Welcome, introductions and course outcomes	Briefing / discussion	Slides 1–15
10:00 AM–11:15 AM	75 min	Topic 1 — Fundamentals of SEO and Generative AI + Lab 1	Visual instruction · demonstration · lab	Slides 16–39
11:15 AM–11:25 AM	10 min	Break	Break	—
11:25 AM–12:30 PM	65 min	Topic 2 — AI-powered keyword research and content strategy + Lab 3	Visual instruction · demonstration · lab	Slides 40–63
12:30 PM–1:30 PM	60 min	Lunch	Lunch	—
1:30 PM–2:45 PM	75 min	Topic 3 — SEO content, images, video and social + Lab 5	Visual instruction · demonstration · lab	Slides 64–87
2:45 PM–2:55 PM	10 min	Break	Break	—
2:55 PM–4:00 PM	65 min	Topic 4 — On-page SEO and Google AI Overviews + Lab 7	Visual instruction · demonstration · lab	Slides 88–111
4:00 PM–4:10 PM	10 min	Break	Break	—
4:10 PM–5:10 PM	60 min	Topic 5 — Performance, ethics and continuous improvement + Lab 9	Visual instruction · demonstration · lab	Slides 112–135

5:10 PM–5:30 PM	20 min	Course recap, Q&A and feedback	Briefing / discussion	Slides 136–143
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Topic and Lab Breakdown

Topic 1 — Fundamentals of SEO and Generative AI • Slides 16–39

Lab 1 — Audit the EcoDesk Demo Website • Slide 37

Establish an evidence-based baseline for a fictional Singapore sustainable-office-products website before any AI-assisted changes.

- Output: Completed baseline audit, five-item priority backlog and preserved original website
- Acceptance: Every finding cites observable evidence, separates fact from inference, includes a verification method and maps to a business or user outcome.

Lab 2 — Design the SIGNAL AI SEO Strategy • Slide 41

Turn the baseline into a 90-day AI-assisted SEO strategy with outcomes, audiences, content priorities, controls and measurement.

- Output: 90-day SIGNAL strategy and governance page
- Acceptance: The strategy links each initiative to an outcome, source of truth, owner, approval gate, metric and rollback or review trigger.

Topic 2 — AI-Powered Keyword Research and Content Strategy • Slides 40–63

Lab 3 — Generate and Validate a Keyword Universe • Slide 61

Create a grounded keyword universe for EcoDesk from supplied first-party notes and synthetic planner/Search Console exports.

- Output: Validated keyword universe and topic-hub webpage
- Acceptance: No metric is AI-invented; every retained keyword has a source, intent, cluster candidate, validation status and business-fit rationale.

Lab 4 — Cluster Keywords and Build Content Briefs • Slide 65

Transform the validated keyword universe into a non-cannibalising topic cluster and three grounded content briefs.

- Output: Topic-cluster architecture and three approved content briefs
- Acceptance: Every keyword maps to one primary page, each brief has a distinct intent and information advantage, and internal links form a logical user journey.

Topic 3 — Creating SEO-Optimised Content with Generative AI • Slides 64–87

Lab 5 — Create a People-First SEO Blog Article • Slide 85

Use a staged prompt workflow to create an answer-first article from a verified brief and claim ledger.

- Output: Verified, edited SEO blog article with claim ledger
- Acceptance: Every material claim is supported or qualified, the page resolves intent, adds original operational value and passes the six-part AI content QA scorecard.

Lab 6 — Generate Image, Video and LinkedIn Assets • Slide 89

Prompt text-to-image and video concepts, write accessible metadata, and repurpose the blog into a LinkedIn post designed to earn attention and editorial backlinks.

- Output: Multimodal SEO asset pack and ethical LinkedIn distribution post
- Acceptance: Assets are relevant, accessible, rights-safe and evidence-aligned; the LinkedIn post offers standalone value and seeks earned rather than manipulated links.

Topic 4 — AI for On-Page SEO and Content Optimisation • Slides 94–117

Lab 7 — Generate and Implement On-Page SEO Meta • Slide 109

Generate metadata options from visible page facts, then implement HTML, internal links, image attributes and valid structured data.

- Output: Optimised static product page and metadata decision log
- Acceptance: Metadata is unique and accurate, visible content and JSON-LD agree, links work, images are accessible and no unsupported rating, price or guarantee is present.

Lab 8 — Optimise an Answer Page for Google AI Overviews • Slide 113

Create a clear, source-grounded page that can be indexed and understood by users and generative search features without relying on unsupported GEO hacks.

- Output: Answer-first guide and AI-feature eligibility checklist
- Acceptance: The page is indexable, resolves the main and follow-up intents with unique value, cites evidence and makes no guarantee of AI Overview inclusion.

Topic 5 — SEO Performance, Ethics and Continuous Improvement with AI • Slides 118–152

Lab 9 — Analyse Search Console and GA4 Data with AI • Slide 133

Use synthetic Search Console and GA4 exports to diagnose performance without confusing impressions, clicks, sessions, users or conversions.

- Output: Evidence-backed SEO performance review and experiment plan
- Acceptance: Every conclusion traces to supplied data, metric scopes are correct, alternative explanations are considered and the recommended test has a baseline and decision rule.

Lab 10 — Build a Governed Agentic SEO Workflow • Slide 137

Integrate the course into a capstone workflow that monitors, recommends and prepares changes while humans retain control of publishing and outreach.

- Output: Governed agentic SEO capstone and monitoring plan
- Acceptance: The workflow is evidence-bound, read-only by default, blocks high-risk autonomous actions, records decisions and demonstrates a successful stop/rollback tabletop.

Resources Required

- Laptop with modern browser and text editor
- Python 3 for the local static web server
- Course slide deck and Learner Guide

- One shared EcoDesk HTML/CSS/JavaScript website plus ten individual activity folders with prompts, tips, articles and synthetic data
- Internet access for the live EcoDesk demo and official source verification
- Open book: course slides, Learner Guide and approved lab materials only.

LMS: <https://lms-tms.tertiaryinfotech.com/>